

This proposal originally dealt with the task of entity typing because we were not provided with enough information. However, after communicating with the team leads, we received clarification that our assigned task deals with a norm-acceptability classification with and without rule-of-thumb problem. Given this new information, we have revised this proposal because now we have a dataset, a more well-scoped problem, and our project now aligns with the original lab project.

1. Title

Knowledge Graph Construction for Cultural Contextualization: Comparing Machine Learning Models for Cultural Norm-Acceptability Classification on NormAd

2. Abstract

Large Language Models (LLMs) have often been shown to fail at representing diverse cultural contexts due to bias in favor of a default, averaged set of values [1]. One solution is to construct a cultural knowledge graph, as proposed in the “Knowledge Graph Construction for Cultural Contextualization” project from Lee Language Lab. This project tests whether automatically generated cultural knowledge graphs help LLMs judge social norms across cultures. Motivated by this, our project focuses on the question: How do course-level machine learning baselines and LLMs compare at classifying whether a behavior is socially acceptable across cultures, and how does running the LLMs with vs. without contextual rule-of-thumb information in the prompt affect accuracy?

Using the provided dataset [akhilayerukola/NormAd](#) from Hugging Face, we will do runs using course-level baseline such as TF-IDF combined with logistic regression or Naive Bayes. We will also do the runs with and without provided rule-of-thumb information which fulfills the sensitivity, ablation analysis requirement. The metric is the macro-F1 score. This baseline performance will then be compared against the LLMs assigned by the lab.

Our findings will provide a benchmark comparing course-level baselines against more advanced models for the task of norm-acceptability classification, which will help with decisions regarding which model to use. Overall, it contributes to the broader study topic of LLMs and cultural representation and inclusion.

3. Research Background

Large Language Models (LLMs) are becoming increasingly prevalent across diverse global populations. However, current research shows that they often homogenize moral diversity, creating issues such as bias, insensitivity, and misunderstanding [1]. As proposed in the Lee Language Lab project “Knowledge Graph Construction for Cultural Contextualization,” one way to address this is to construct cultural knowledge graphs to provide LLMs with structured cultural context. The original lab project focuses on using this to align LLMs to reflect diverse human values and norms across cultures. Therefore, we were assigned a number of LLMs to evaluate on NormAd-ETI, a dataset of about 2,600 short stories set in 75 countries, each asking whether a behaviour is socially acceptable (yes, no or neither). Then, their experiments will add knowledge-graph context to the prompt and measure the change.

This problem is worth investigating because a controlled comparison of advanced models and simple baselines helps identifying the most robust and efficient methods for the task of norm-acceptability classification.

4. Methodology

We will use the provided dataset akhilayerukola/NormAd from Hugging Face. The task is a 3-class classification (yes, no, neither) of whether a behavior is acceptable across cultures. We will collect data from runs on a course-level baseline (TF-IDF + logistic regression or Naive Bayes) to meet the requirement. These results will then be compared against the assigned LLMs InternLM-2.5 7B and Gemma-2 27B (with and without rule-of-thumb for each LLM). Our evaluation metric will be macro-F1 to treat the three classes (yes, no, neither) equally.

5. Significance

This project provides a comparison of how different machine learning methods perform at equitable cultural alignment. Our findings can reveal if simple course-level methods suffice to help determining how much computational complexity is worth it and also if providing rule-of-thumbs help LLMs at this

task.

6. SMART Timeline

We have a timeline in the team contract.

7. References

[1] S. Munker, “Cultural Bias in Large Language Models: Evaluating AI Agents through Moral Questionnaires.” July 14, 2025. Accessed: Sept. 19, 2026. [Online]. Available: <https://arxiv.org/abs/2507.10073v2>